

Jennifer Paige

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Education

- **University of California, Davis**, 2022-current
 - PhD Student in Graduate Group in Applied Mathematics, advised by Alan Hastings
 - DOE Computational Sciences Graduate Fellow 2023
 - NSF National Research Traineeship: Sustainable Oceans Trainee, Cohort 5
- **Swarthmore College**, Pennsylvania, 2018 -2022
 - Bachelors of Arts in Mathematics & Educational Studies
 - Sigma Xi, AWM, and MAA member

Skills & Relevant Coursework

- Coursework in differential equations, asymptotics, numerics, linear algebra, and analysis
- Coding experience in Python, R, Matlab, and SageMath; proficient in LaTeX

Employment

- **2023: Graduate Research Student at Cawthron Institute, New Zealand**
 - Developed mathematical models related to optimizing spatial marine management and risk assessment, through application of particle tracking software Oceantracker
- **2021: Yale SUMRY Research Experience for Undergraduates student** - Diffusion Geometry & Topology group
 - Worked to harness geometric information, specifically curvature, about point clouds to inform data analysis methods such as topological data analysis
 - Gained background in topological data analysis, geometry, and computer science, as well as further developed technical writing and research skills (independently and in a team)
- **2021-22: Biology Student Research Assistant at Swarthmore College**
 - Studying aging and genetic diversity in asexually reproducing planaria worms through the analysis of reproductive data
 - Monitored and imaged a population of *Schmidtea Mediterranea*
- **2020-22: Student employment in the Mathematics Department at Swarthmore College**
 - Math “Clinician” Peer Assistant - assisting all levels of courses by hosting help sessions
 - Math “Pi-rate” Peer assistant - assisting introductory calculus classes through in-class assistance and help sessions
 - Mathematics Grader - graded for Differential Equations course
- **2020: Gerrymandering Research Experience for Undergraduates student**
 - Used random walks and polygon maps to study the level of gerrymandering present in states
 - Gained mathematical and coding skills through developed programs that apply Markov chain Monte Carlo algorithms

- 2019-20: **Educational Psychology Research Assistant at Swarthmore College**
 - Studied how to improve Bootstrap program (teacher training program for the integration of math and computer science into classrooms) across the country through survey data analysis
- 2016-20: **High School and Undergraduate Research Student at Los Alamos National Laboratory Theoretical Physics (XTD-PRI) (2018-20)**
 - Analytically solving of hydrodynamic mechanics and symmetry analysis problems
- **Computational Earth and Environmental Sciences (EES-16) (2016-18)**
 - Developed soil heat transfer model in Python and Fortran and visualizations in Paraview
 - Analyzed moisture sensitivity of fire simulator (FIRETEC)
 - Assisted in the development of advanced turbulence equations for small scale perturbations

Papers

- Bhaskar, D. et al. "Diffusion-based methods for estimating curvature in data." ICLR 2022 Workshop on Geometrical and Topological Representation Learning, Virtual, Apr. 2022. openreview.net/pdf
- Paige, Jennifer Nicole. "Using Differential Forms to Find Symmetries in the Noh Problem for an Ideal Gas in a Spherical System." Los Alamos National Laboratory. 28 Jan. 2020. [LA-UR-20-20859](https://www.osti.gov/servlets/purl/1570599).
- Davis, Diana et al. "Assessing Congressional Districting in Maine and New Hampshire." 12 Nov 2020. [arXiv:2011.06555v1](https://arxiv.org/abs/2011.06555v1).
- Renninger, K. Ann et al. "Collaborative Research: Hybrid Professional Development to Enhance Teachers' Use of Bootstrap." [Under review]. Educational Studies, Swarthmore College.

Conferences & Presentations

- Invited participant at Geometric and Topological Methods in Data Science Conference, ICERM, Dec. 2021.
- Paige, J., MacDonald, K., Thomas, D., Zhao, S. "Towards Robust Curvature Computation in Point Clouds." University of Connecticut's REU Vir(tu)al Conference. Aug. 2021.
- Paige, J., MacDonald, K., Thomas, D., Zhao, S. "Towards Robust Curvature Computation in Point Clouds." Yale's SUMRY 2021 Weekly Program Symposia. July-Aug. 2021.
- Holland, Troy et al. "A Case Study of Wildfire/Atmosphere Coupling on Complex Topography." Los Alamos National Laboratory. 11 May 2018. Presentation. LA-UR-18-24156.

Leadership & Community Engagement

- 2022 Heinrich W. Brinkmann Mathematics Prize recipient- for outstanding performance in the field and exemplary services to the Department
- 2020-21, 2021-22 Board member of Swarthmore Gender Minorities in Math & Statistics
- 2021-22 Math and Statistics Student Advisory Council member
- Mathematics tutor for Chester Dare 2 Soar program